(5.2) Radian Measure and Angles on the Cartesian Plane

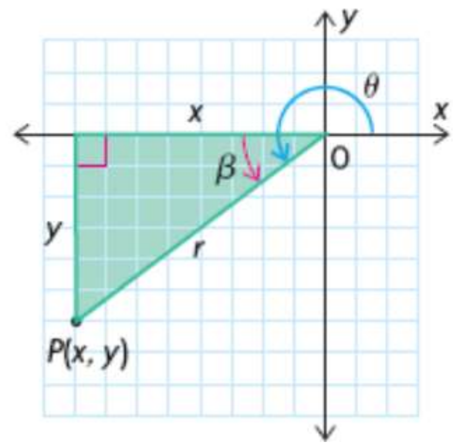
Recall:

$$\sin\theta = \frac{y}{r} \qquad \csc\theta = \frac{1}{\sin\theta} \qquad \csc\theta = \frac{r}{y}$$

$$\cos\theta = \frac{x}{r} \qquad \sec\theta = \frac{1}{\cos\theta} \qquad \sec\theta = \frac{r}{x}$$

$$\tan\theta = \frac{y}{x} \qquad \cot\theta = \frac{1}{\tan\theta} \qquad \cot\theta = \frac{x}{y}$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$



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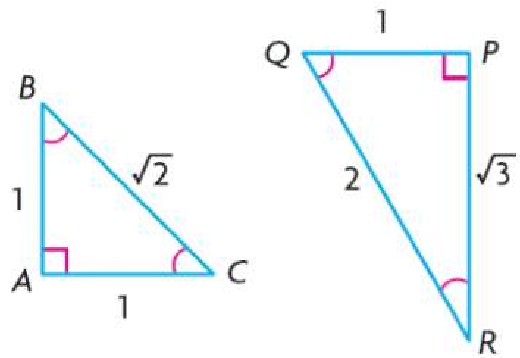
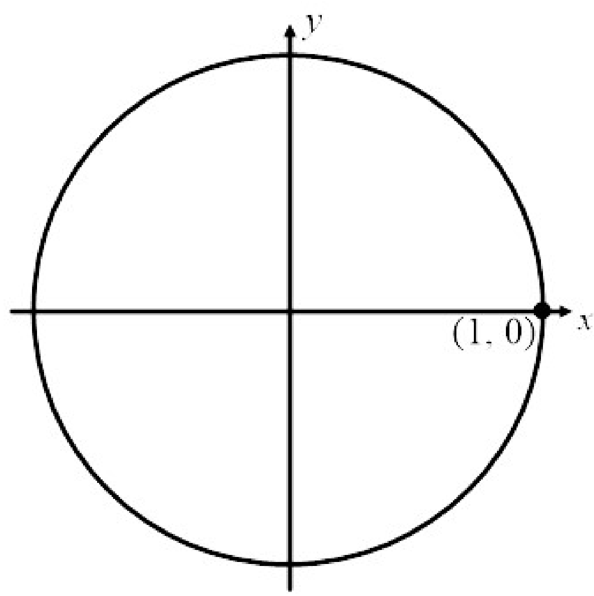


We use the Unit Circle, Special Triangles, the CAST Rule, principal angles, and related acute angles to understand different ratios of trigonometric values:



Ex. 1 Determine the exact value of each:

(a) $\sin\frac{\pi}{4}$

(b) $\tan\frac{7\pi}{6}$

(c) $\csc\frac{3\pi}{4}$

(d) $\cot\frac{3\pi}{2}$

Ex. 2 If $\cos\theta = -\frac{5}{13}$ where $0 \leq \theta \leq 2\pi$, evaluate θ to the nearest hundredth.

Ex. 3 State $\cos\left(\frac{4\pi}{3}\right)$ as an equivalent expression in terms of the related acute angle.